

enabled needle positioning feedback in real time in case of interventional procedures.

In addition, nowadays, ultrasonic images of an area are fused with the images of other modalities, like CT and MRI images of the same areas, to further enhance the diagnosis experience.

Image super-resolution

On first observation of the ultrasonic image in Fig. 2, it can be seen that ultrasonic images suffer low spatial resolution due to weak received echo signal from the target reflectors, along with a combination of Gaussian and Speckle noise. These drawbacks can be hard to address through any hardware changes.

Apart from these demerits, ultrasonic images have very low resolution details as compared to CT and MRI. When any hardware changes are attempted, the equipment can become expensive and may pose some other unforeseen problems. An inexpensive and effective solution is the correction of images through software; if it poses any problem the system can be easily modified as per requirement.

In this article, we propose a super-resolution method to enhance the quality and hence visual perceptiveness of the ultrasonic images by developing a hybrid approach. Flow chart for the proposed method is shown in Fig. 2.

We start with a low-spatial-resolution ultrasonic image and apply non-local dehazing to it as the input image appears to be a little bit hazy and blurry. Due to this, there is very less visibility and contrast; also there is lack of proper luminance.

The non-local dehazing method is a linear, non-trained method that efficiently recovers the true characteristic image from the hazy one. It also handles the noise with care and makes the contrast better.

Fig. 2 shows the result after non-local dehazing. The resulting image has good contrast and better luminance. The details are revealed further when we remove the noise using the Guided filter. Guided filter is an edge preserving smoothing filter that effectively removes the noise while preserving the edges with high contrast details. It is an explicit image filter that uses a reference image known as the

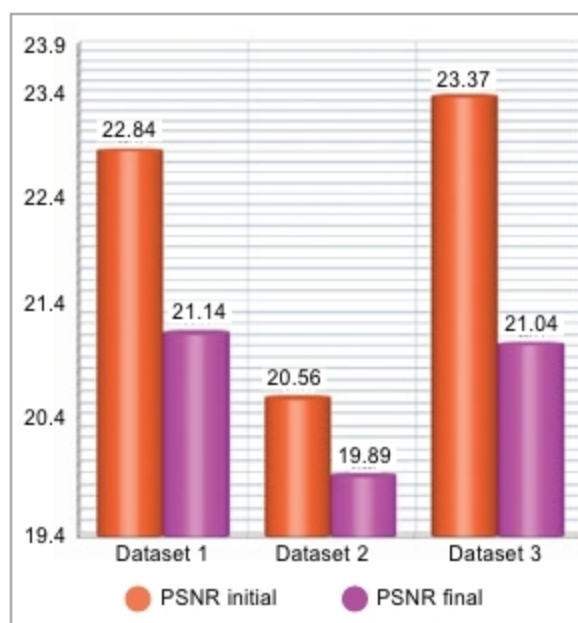


Fig. 6: Initial and final PSNR for the three datasets

guidance image to look for the edges to be preserved.

In the proposed method, a square window of 16 is chosen to make operation fast and the edge selectivity controlling parameter of 0.0016 is chosen. The edge selectivity parameter is chosen to be low because for large values of the parameter the filter converges to a normal box filter with lost edge preserving properties. The guided filter brings out subtle details and makes them more visible as seen in Fig. 2.

It is well known that colour images are more visually appealing than grayscale images. The additional colour component in the images helps the human perceptual system to perceive better. So a red blood colour is injected into the filtered image to make it more visually appealing. The final image looks very informative and gives high and accurate spatial as well as temporal resolution.

This method is applied on three different datasets and objective evaluation of the method is done on the basis of peak signal-to-noise ratio (PSNR), which calculates overall signal strength over the noise strength and mean square error (MSE) that calculates the distance of corresponding pixels between final image and reference image (input image). The visual results of the proposed method on three different datasets are shown in Figs 3, 4 and 5. The objective evaluation results are shown in Figs 6 and 7.

Although the objective evaluation results show decrease in PSNR and increase in MSE, the visual results have improved a lot, which is evident

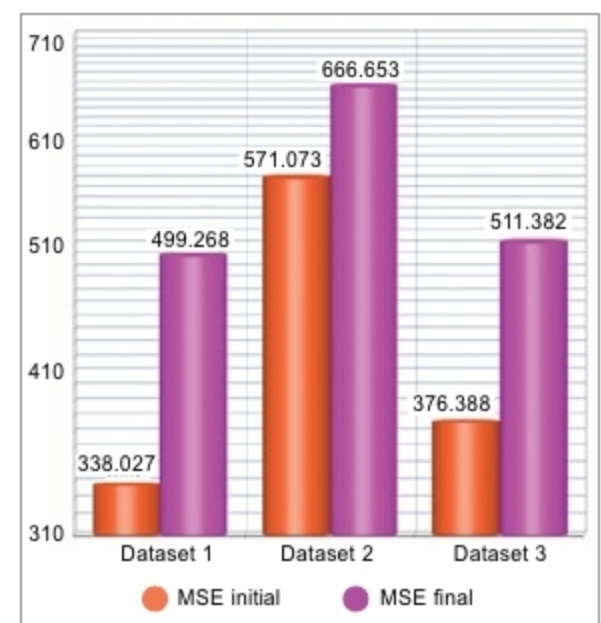


Fig. 7: Initial and final MSE for three datasets

from the final results obtained. The purpose of providing super-resolution to ultrasonic images is achieved through this proposed architecture, which can be applied seamlessly to any ultrasonic image without adding any extra cost of implementation.

It is clearly visible that the proposed architecture has a strong prospect of extending the efficacy of ultrasound images with high spatial and temporal resolutions. The method clearly retrieves the hidden details of the common carotid artery.

Conclusion and future

It is well known that CT and MRI imaging have a very distinctive place in the field of medical imaging and diagnosis. These images have much better resolution and contrast, and provide comparatively more information content about the target areas than ultrasonic images, but these are also much costlier than ultrasonic imaging.

A simple and cost-effective solution to this is ultrasonic imaging. It does not provide as good image as a CT or MRI, but with each passing day and with technology getting better and better, ultrasonic imaging systems are also becoming better.

It may not be wrong to think that, focusing primarily on ultrasonic imaging and making it better through various means will help medical fraternity to reach more widespread areas where bringing other imaging technologies is difficult. This article is one such effort to improve ultrasonic imaging at the software end, so that imaging quality can be improved at a fraction of cost. **EFY**